

Quiz (Carolina Reaper)

- Q1.** What is the graph of the system $x + y < 2$ and $x - y > -1$?
- (A) Only the 1st quadrant
(B) Only the 2nd quadrant
(C) Only the 3rd quadrant
(D) Only the 4th quadrant
(E) 2nd and 4th quadrants
- Q2.** Which of the following is a solution to the inequality $x + 2y > 4$?
- (A) (1, 1)
(B) (2, 1)
(C) (1, 2)
(D) (0, 0)
(E) (2, 2)
- Q3.** The graph of $x > 2$ and $y < 3$ is
- (A) a line
(B) a parabola
(C) a circle
(D) the 4th quadrant
(E) the entire coordinate plane
- Q4.** The system of inequalities $x > 1$, $y > 1$, and $x + y < 5$ has a solution region that is
- (A) a triangle
(B) a square
(C) a circle
(D) a rectangle
(E) the entire coordinate plane
- Q5.** What is the solution to the system $y > x - 2$ and $y < x + 2$?
- (A) the line $y = x$
(B) the line $y = -x$
(C) the entire coordinate plane
(D) only the y-axis
(E) only the x-axis
- Q6.** The feasible region of the system $x + y > 2$ and $x - y > 0$ is
- (A) the 1st quadrant
(B) the 2nd quadrant
(C) the 3rd quadrant
(D) the 4th quadrant
(E) the entire coordinate plane
- Q7.** The system of inequalities $y > x$ and $y < -x$ has a solution region that is
- (A) the 1st and 2nd quadrants
(B) the 2nd and 3rd quadrants
(C) the 3rd and 4th quadrants
(D) the 1st and 4th quadrants
(E) the entire coordinate plane
- Q8.** The graph of $y > -x + 2$ and $y < x + 2$ is
- (A) two parallel lines
(B) two intersecting lines
(C) the x-axis
(D) the y-axis
(E) the entire coordinate plane

Open Ended Questions (FRQ)

- Q9.** Graph the system $x + y > 1$ and $x - y < 3$, labeling all axes and identifying the solution region.
- Q10.** The system $y > x - 1$ and $y < -x + 3$ has a solution region that is a triangle. What are the coordinates of the vertices of the triangle?

Chef's Tips

- Provide clear instructions and examples before the quiz.
- Encourage students to ask questions if they are unsure about any part of the quiz.
- Remind students to label their axes and provide units where applicable.